SADE-SL Low-Cost Net Contract Statement of Work

Section C - Descriptions and Specifications

STATEMENT OF WORK

C. DESCRIPTIONS AND SPECIFICATIONS for Sade-SL Low-Cost Net

C.1. Scope: This Statement of Work (SOW) describes the design, test, fabrication, performance requirements, and deliverables required for the SADE-SL Low-Cost Net effort.

C.1.1 System Description: SADE-SL consists of four main capabilities (Payload Stabilization, Enhanced Speed Bag, Low-Cost Cargo Net, and Low-Cost Sling Sets) which provide options for the soldiers to provide distributed supply and sustainment support. The two low-cost capabilities (low-cost slings sets, low-cost cargo nets) reduce the overall cost of sling load equipment and the burden of intra-theatre recovery. The other two capabilities allow for improved flight safety in the form of increased air speed by 10%-20% (Payload Stabilization Device) and decreasing hovering time to 14 seconds from time on target (Enhanced Speed Bags) as well as a 92% up to 95% survivability of the payload.

For this contract the goal will specifically be to design the Low-Cost Cargo Net capability of SADE-SL. This will need to be created to support the 5,000lb and 10,000lb payload weight classes and will need to match the physical characteristic requirements of the legacy 5,000lb and 10,000lb Cargo Nets. Most importantly the Low-Cost Cargo Net must be at a 35%(T) and 75%(O) reduction in overall item cost from the legacy cargo net for both weight classes.

SADE-SL Low-Cost Cargo Net must be compatible with UH-60 and CH-47 aircraft by maintaining all existing interfaces between the aircraft and apex fitting of the Low-Cost Cargo Net.

C.2 Applicable Documents

DODI 5000.88 Section 3.5a Engineering of Defense Systems
MIL-STD-31000A, Department of Defense Standard Practice: Technical Data Packages
DI-MISC-80711, Data Item Description: Scientific and Technical Reports
DI-QCIC-81722, DI-QCIC-81722, Data Item Description: Quality Program Plan. (QPP)
MIL-STD-810G, DoD Test Method Standard: Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1472G, DoD Design Criteria Standard: Human Engineering
MIL-STD-1366E, DoD Interface Standard for Transportability Criteria.
MIL-STD-2073-1E, Standard Practice for Military Packaging.
MIL-STD-913A, Requirements for the Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopters
STANAG 2286 (Edition 2), Technical Criteria for External Cargo Carrying Slings, Nets and Straps/Pendants

C.3 Acronyms and Definitions

(T) - This refers to a threshold/ minimum a requirement needs to meet to be acceptable.

(O) - This refers to an objective or best-case value for a requirement.

HSL – Helicopter Sling Load

SPT – Static Proof Testing

C.4 Requirements

C.4.1 General Requirements for SADE-SL:
The requirements for the SADE-SL Low-Cost Cargo Net design and new Low-Cost Cargo Net components are the following:

- Must be able to support the suspended weight capability in the standard system categories of 5,000 and 10,000 pound capacity at a 35% (T) and 75% or more (O) reduction of individual item cost compared to the current nylon cargo nets.
- The SADE-SL components shall not increase the current sling load equipment rigging time to rig the load by more than 25% (T) and reduce the load rigging time and/or number of personnel required by 25% or more (O) Allow for ease of rigging for future variants of payloads with minimal redesign, test and certification, as well as reduce impacts on the environment.
- SADE-SL shall be compatible to current Army rotary wing aircraft that include the UH-60 and CH-47 (T) and compatible to other military and civilian rotary wing aircraft to include the CH-46, CH-53 and CV-22, Future Vertical Lift (FVL) medium and heavy lift aircraft (O)
- SADE-SL shall be compatible to standard or recently developed sling load and airdrop components to the maximum extent possible (T=O).
- The SADE-SL components shall be capable of operation and storage in all of the following climatic categories; Basic, Hot and Cold as current sling load and aerial delivery equipment/systems (T=O). Function in Basic, Hot and Cold conditions in accordance with MIL-STD-810G. These variables will be hot, basic, cold, ranges as low as -60o F (objective) to +145o F and high-altitude pressure standards characteristic of worldwide operations, as described in AR 70-38, dated 15 Sep 1979.
- It will operate within all terrain types (desert, mountain, jungle, and water/maritime), and during inclement weather, to include precipitation, air carrier Instrument Meteorological (IMC) conditions and maritime/salt fog conditions.
- As required by the Government, compliance may be verified at a Government approved facility in accordance with MIL-STD-810G, Method 500.4 (Low Pressure/High Altitude, and Rapid Decompression), Method 501.4 (High Temperature), Method 502.4 (Low Temperature), Method 505.5 (Solar Radiation – Sunshine Procedures 1,2), Method 506.4 (Rain Drip), Method 507.4 (Humidity), Method 508.4 (Fungus), Method 509.4 (Salt Fog), Method 510.4 (Blowing Dust), Method 510.4 (Blowing Sand), Method 511.4 (Explosive Atmosphere), Method 514.5 (UH-60, CH-47, C-130, C-17 and Loose Cargo Vibration).
- Capable of adaptation to other US forces for strategic, operational, and tactical military operations. Shall be designed to promote ease of operation and maintenance by the 5th percentile female through the 95th percentile male Soldier/Marine/Airman as described in MIL-STD-1472G while wearing the field duty uniform, cold weather uniform, and chemical protective uniform.
- Must demonstrate suitability in conditions where the user wears cold weather gloves. The system shall be capable of being transported by highway, rail, marine, and air modes, world-wide. The applicable transportability criteria are set forth in MIL-STD-1366E.

C.4.2. SADE-SL Low-Cost Cargo Net(s) development requirements:

The contractor shall develop the SADE-SL Low-Cost Cargo Net system in the following areas.

C.4.2.1 SADE-SL Low-Cost Cargo Net Design: The contractor will develop the SADE-SL Low-Cost Cargo Net system to provide a balance of performance, compatibility with rotary wing aircraft, decreased or not increased by 25% of rigging time, and decreased cost. The Low-Cost Cargo Nets must provide effective payload support across the full range of payload weights, from 5,000 – 10,000 pounds rigged weight.

C.4.2.1.1 System Design Validation and Preliminary Design Review (PDR)
To execute this section of the SOW, the contractor must fabricate and deliver test prototypes to the government, as described in section C.4.6, to be used in static proof testing.

SADE-SL Low-Cost Net Static Proof Testing (SPT) will be conducted by the government at the Natick Soldier Systems Center. The contractor must include a proposed plan for SPT, to be executed by the government.

Two test weeks are planned for the Low-Cost Cargo Net to accomplish SPT of the system. SPT will be conducted up to the maximum payload weight limits. A first iteration of SPT may be used to test variations in system design and gather data which could improve the design and system performance. The contractor will use these test results to improve the performance and reliability of the design, and update prototypes if necessary. The second event will serve to test any additional changes and increase confidence in the design performance and reliability. The contractor is required to be on-site for SPT testing to measure system performance and to evaluate any damage to the Low-Cost Cargo Net prototype. The contractor is required to provide sensors, data acquisition and analysis during SPT. If the Low-Cost Cargo Net fails to meet test performance requirements or there is a Cargo Net equipment failure, the contractor is required to document the issue and conduct a failure analysis. The contractor will then propose design changes to improve the performance or prevent the failure mode and perform the corrective action. If failure analysis of the Low-Cost Cargo Net is required, it will be shipped to the contractor facility by the government. The contractor will then deliver a redesigned net to the government for a re-test of the SPT.

The purpose of the Preliminary Design Review is to ensure that the prototype design is sufficiently complete, and test results provide confidence the performance, durability, and reliability requirements of the system will be satisfied. The design at PDR must show compliance with HSL requirements as specified in MIL-STD-913. In accordance with CDRL A001, the contractor will deliver a PDR technical reporting package which describes the prototype system design, specifications, test data and analysis, and must demonstrate a Technology Readiness Level (TRL) 6. The contractor must provide technical direction for any new procedures required for SADE-SL Low-Cost Net equipment operation. The presentation format and content detail will be of the contractor's choosing but must be sufficient to support the purpose of the PDR. A successful PDR will allow the government to approve the SADE-SL Low-Cost Net design for flight testing and the continuation of prototype fabrication.

C.4.2.1.4 Helicopter Sling Load (HSL) Design Validation (DV) Testing:

To execute this section of the SOW, the contractor must fabricate and deliver test prototypes to the government to be used in SPT as described in section C.4.6.

SADE-SL Low-Cost Net Flight Testing is conducted by the government. The contractor shall provide test support in accordance with the SOW section C.4.3.

The initial phase of airdrop flight testing will be conducted for Design Validation (DV) of the prototype design approved at the PDR. The initial DV airdrop phase may utilize ballast or “Weight tub” payloads. These tests are conducted to show the performance of the Low-Cost Net system without the dynamic interaction of actual payloads. The results from these tests will be used to validate the system performance predictions. Applicable payloads will be included in the DV flight test phase to demonstrate proof of concept and development of rigging procedures. The goal of Design Validation testing is to stabilize the product baseline to be approved and baselined at the Critical Design Review.

C.4.2.1.5 Critical Design Review (CDR).

The Critical Design Review is conducted by the government at the end of the Design Validation (DV) phase to establish the system’s initial product baseline. The purpose of the CDR is to confirm that the system design is stable and is expected to meet system performance and reliability requirements. In accordance with CDRL A002, the contractor will deliver a technical reporting package for CDR which details the system design, specifications, test data and analysis. The system design maturity and test results must demonstrate a Technology Readiness Level (TRL) 7. A successful Critical Design Review will provide the contractor approval to proceed with fabrication of additional prototypes for developmental testing.

C.4.2.1.6 Flight Developmental Testing (DT).

To execute this section of the SOW, the contractor must fabricate and deliver test prototypes to the government to be used in flight testing as described in section C.4.6.

SADE-SL Low-Cost Net flight testing is conducted by the government. The contractor shall provide test support in accordance with the SOW section C.4.3.

The objective of developmental flight testing (DT) is to demonstrate performance and reliability across the planned operational range of the system. DT is conducted with the prototype design approved at the CDR. Any design changes during DT must be formally documented and approved by the government and are primarily limited to addressing minor issues with the system. Operational payloads will be included in the DT airdrop phase to demonstrate performance of the SADE-SL Low-Cost Net in delivering mission capable payloads. The objective of DT is to verify the system performance and reliability will meet requirements.

C.4.3 Engineering support and data analysis for Flight Testing.

To execute this section of the SOW, the contractor must fabricate and deliver test prototypes for flight testing as described in section C.4.6.

The contractor will be required to be onsite for SADE-SL Low-Cost Net flight testing or additional SPT events. Locations for test support are Aberdeen Proving Grounds, Md; and Natick Soldier Systems Center, MA. The contractor must complete the following requirements for each test event:

Pre-Test Preparation:
- Test Coordination/Readiness Meeting
- Review of the planned test configuration and flight test parameters.
- Preparation of SADE-SL Low-Cost Net developmental prototypes.

During Test Week:
- Equipment inspection on drop zone.
- Failure analysis during test event
- Post-Test Analysis and Corrective Action
- Conduct an analysis of system performance using government data.
- Unexpected results and test abnormalities should be investigated through review of all data including videos and photos.
- Failure analysis as required and proposal of corrective action.
- If failure is attributed to deficiencies in developmental design, complete corrective action.
- The contractor shall submit a test report with the above data and any additional information in accordance with CDRL A003.

Due to the one time use requirement for SADE-SL Low-Cost Net during testing, the contractor will be required to ensure appropriate number of prototypes are supplied for testing.

C.4.4 Logistics Support
The contractor is required to provide the following support for the government's development of SADE-SL Low-Cost Net logistics.

C.4.4.1 Physical Configuration Audit - The contractor shall conduct a Physical Configuration Audit (PCA) on the final SADE-SL Low-Cost Net prototype design. This will be a complete disassembly of the SADE-SL Low-Cost Net(s) and inspection for compliance with the Technical Data Package (section C.4.5.3). The PCA will occur over the course of 1 week at the contractor's facility.

C.4.4.2 Packaging Design:
The contractor shall design and test packaging for all SADE-SL Low-Cost Net item(s) classified as Special Group according to MIL-STD-2073-1E.

Items will be classified as special group of they have a unit pack weight exceeding 40 pounds and or a dimension greater than 40 inches. In addition, the item will be classified as special group if the unit pack length and girth combined exceed 84 inches. Special group items often require sketches, figures, or narrative instructions to describe packaging requirements. Items excluded from the Selective group will be classified as Special group items. This includes kits, sets and items of separate parts, items requiring disassembly, repairable items, items requiring special handling or condemnation procedures, items classified as hazardous material or hazardous goods in transport, items assigned a shelf life, electrostatic discharge sensitive items, fragile, sensitive, and critical items.

The contractor shall deliver the packaging design to include specification of all parts (boxes, skids, cartons), materials and construction of special boxes. Packaging design shall be in accordance with MIL-STD-2073-1E, Method 10 - Physical protection. The unpreserved item(s) shall be protected from physical damage and mechanical malfunction. Cushioning materials, dunnage, blocking and bracing shall be applied as required to protect the item(s) and the enclosing media and restrict the movement of the item within the container. The materials selected for all blocking and bracing, and the design and application thereof, shall be compatible with the item and its bearing load limitations. The packaging design must be sufficient to pass Validation testing of Special group items in accordance with ASTM D 4169 (Standard Practice for Performance Testing of Shipping Containers and Systems) Distribution Cycle 18, Assurance Level I, with Acceptance Criterion 3 (product is damage free and packaging is intact). Validation testing may be limited to Test Schedule 1 and Test Schedule 6 for Distribution Cycle 18. Replicate testing and climatic conditioning will not be required during testing. The contractor is not responsible for conducting the testing.

C.4.4.2.1 Package Configuration Management:
The contractor shall deliver a packaging design for all configurations of the SADE-SL Low-Cost Net(s), including up to two configurations for the 5k and 10k weight requirements. The contractor shall submit the packaging design in accordance with CDRL A004. If the packaging fails any verification tests, the contractor must redesign the packaging and so the government can conduct a re-test. The contractor is not required to deliver packaged equipment for testing under this section.

C.4.5. SADE-SL Low-Cost Net technical data package:
All design changes and SADE-SL Low-Cost Net configurations must be documented in an update to the Technical Data Package (TDP). The Contractor's final TDP delivery must be in accordance with MIL-STD-31000A, dated 26 Feb 2013. The TDP includes product drawings, associated parts lists, and 3D Computer Aided Design (CAD) models which must provide the design, engineering, manufacturing, and quality assurance requirements information necessary to enable the procurement or manufacture of an item essentially identical to the original item. The product shall be defined to the extent necessary for a competent manufacturer to produce an item, which duplicates the physical, interface, and functional characteristics of the original product, without additional design engineering effort or recourse to the current design activity. Product data shall reflect the approved, tested, and accepted configuration of the defined delivered item.

C.4.5.1 Draft TDP:
The contractor shall submit a Draft TDP for the SADE-SL Low-Cost Net, in accordance with CDRL A005.

C.4.5.2 Final TDP:
The contractor shall submit a Final TDP for the SADE-SL Low-Cost Net Configuration(s), in accordance with CDRL A005.

C.4.6. Fabrication of SADE-SL Low-Cost Net Prototypes:
The contractor shall fabricate and deliver prototype equipment required for SADE-SL Low-Cost Net development and testing. This includes the Apex Fitting, four Apex Fitting Hooks (for each leg) and up to two configurations to meet the performance for a 5k/10k Net

C.4.6.1 Fabricate and Build SADE-SL Low-Cost Net assemblies:

The contractor shall fabricate and deliver SADE-SL Low-Cost Net prototype assemblies. The Low-Cost Net assemblies will be delivered in accordance with the approved TDP and design changes approved by the government through the work described in Section C.4.2.1.

C.4.6.2 TM Validation and LUE Events:
The contractor shall support logistics events including the TM Validation and Limited User Evaluation.

C.4.7. Quality Control

The Contractor shall develop and implement a Quality Plan in accordance with CDRL A006 that outlines the quality standards and requirements for the manufacturing process. This plan shall include procedures for inspecting and testing the manufactured parts and assemblies at various stages of production, as well as guidelines for identifying and correcting any non-conformities. The Quality Plan shall only address the production and delivery of prototypes and does not apply to other engineering and development that are included in this statement of work.

C.4.7.1 The Contractor shall perform regular inspections and tests on all manufactured parts and assemblies during the manufacturing process to ensure compliance with all applicable specifications and requirements. The Contractor shall also document all inspection and test results as specified in the Quality Plan.

C.4.7.2 The Contractor shall implement a non-conformance management process that includes procedures for identifying, documenting, and correcting any non-conformities identified during the manufacturing process. The Contractor shall also track and report on all non-conformities, as well as any corrective actions taken.

C.4.7.3 The Government reserves the right to request inspection reports and material certifications from the Contractor for the lot based upon sampling per ANSI/ASQ Z1.4. When certificates of conformance (CoC) for materials and/or end items are required, testing data documentation must accompany the CoC.

C.4.8. Configuration Management.

The contractor shall implement and maintain an internal Configuration Management (CM) program for the SADE-SL Low-Cost Net, throughout the life of the contract. EIA649, 'National Consensus Standard for Configuration Management', may be used as a guide for the contractor's configuration management program. Copies of this document may be purchased at http://www.sae.org/. At any time, the government shall observe the contractor's CM processes to ensure quality product performance and minimize impacts to the project schedule.

C.4.8.1 Physical Configuration Baseline (PCBL): The contractor shall create and control the PCBL using the change control and engineering release processes. The PCBL, which shall be in the contractor's own format, is the product performance requirement for replacement assemblies and spare/repair parts, engineering drawings, parts lists, process specifications and computer software configuration items. The PCBL shall support interchangeability and interoperability to a replaceable part level.

C.4.8.2 Configuration Control: The contractor shall use configuration control to manage all proposed changes after the initial product baseline is established at CDR (See 4.2.1.5 and 4.2.2.5). Configuration control shall be used to document the impact of proposed changes and to update configuration documentation. Following acceptance of the system, the contractor shall not alter the design in form, fit, or function without prior approval from the government Contracting Officer (KO).

C.4.9. Contractor Responsibilities

C.4.9.1 The contractor shall furnish all personnel, labor, engineering, services, materials, supplies and facilities necessary to design and manufacture the SADE-SL Low-Cost Net system and provide all support hardware as indicated in this Statement of Work (SOW). The work and services to be performed by the contractor are detailed in this SOW and will be authorized by issuance of Delivery Orders (DO). The contractor shall not initiate any work that is not authorized by a DO or modification without written direction by the Contracting Officer.

C.4.9.2 Project Management:

The contractor shall designate an individual as the Contractor's Program Manager (PM). The PM shall serve as the primary Point of Contact (POC) between the government and contractor and shall be responsible for the coordination of all contractor activities related to the contract. The PM shall have the authority to commit the contractor to specific courses of action and accept direction from the Contracting Officer. The PM shall be responsible for coordinating all meetings between the government and the contractor. The PM shall be responsible for bringing to the Contracting Officer's attention any conflicts in the contractor's interpretation of the contract requirements (first by telephone and followed in writing) or problems that could adversely affect the contractor's ability to meet the stated quality, cost, or production/delivery schedule.

C.4.9.3 Project Plan and Schedule:

The contractor shall develop and maintain a project plan and schedule that outlines all the tasks required to execute the program. A baseline plan and schedule shall be established during the contract preparation phase and tracked from contract award. The plan and schedule shall show in detail the path the contractor will follow to meet the required delivery dates of all items awarded on the contract. The contractor is expected to keep the plan and schedule up to date and track program progress using the schedule.

C.4.10. Meetings and Reviews

C.4.10.1 Post Award Conference: The contractor shall coordinate, schedule, and conduct a Post Award Conference with the government within 14 calendar days of Contract award. The purpose of the Conference shall be discussion of project orientation, transfer of background information, to provide a mutual understanding of the technical requirements/contractual requirements and the Quality Assurance provisions of the contract. The government can address any questions or issues with regards to technical matters. The contractor shall describe to the government the management of all aspects of the program. The contractor shall ensure that all personnel and subcontractors that are required for an adequate discussion of the contract effort are in attendance. Scheduling of the Post Award Conference shall not change the delivery schedule of the contract. The contractor shall be prepared to:

- Conduct a review of the system requirements to ensure that they have been completely and properly identified and that there is a mutual understanding of the system requirements between the government and the contractor.
- Make available to government representatives the documentation for production planning, manufacturing methods and controls, material and manpower resource allocation, production engineering, quality control and assurance program, and management of major subcontractors.
- Review the plan, tasks, and schedule required to execute the SADE-SL Low-Cost Net development program within the schedule constraints set forth by the contract/government.
- Document the Post Award Conference meeting minutes and distribute the minutes via e-mail to all Post Award Conference attendees no later than 1 week from the end of the Post Award Conference.

C.4.10.2 Design Review Meetings and Information Packages: The contractor shall participate in design review meetings by presenting the technology readiness review packages described in sections C.4.2. A minimum of 14 calendar days after each review shall be given to the government for providing comments and concerns regarding the design. Government concerns and comments shall be resolved as part of the Contractor's Design Review Information Packages and shall be submitted to the government.

C.4.10.3 Monthly Teleconference Meetings: A monthly Teleconference Meeting shall be conducted by the SADE-SL Integrated Product Team (IPT) of government and contractor personnel to address technical progress (design, development, fabrication), clarify requirements, review quality assurance, cost, schedule, contractual, and other programmatic issues, or concerns. The schedule for Monthly Teleconference meetings will be flexible and occur on an as needed basis as program issues dictate. In addition to the monthly teleconference, regular email and phone or teleconference communication is a requirement for this project to maintain clarity of requirements and objectives and sharing of information.